

Krishnaa Vadivel

Computational Materials Physics | Scientific Computing

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Portfolio: <https://krishnaa423.github.io/personal-website/> | GitHub: <https://github.com/krishnaa423>

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Summary

Computational-physics PhD researcher who turns many-body and first-principles models into scalable scientific software. Combines Python/C++, MPI/OpenMP, CUDA/ROCm, PETSc/SLEPc, and scientific ML for materials calculations on leadership-class GPU systems.

Technical Skills

Scientific computing: Python, C++, MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, DOLFINx, NumPy, SciPy, SymPy

GPU and HPC: CUDA (cuBLAS, cuTENSOR), ROCm (rocBLAS, rocFFT), CUDA-aware MPI, collective communication, strong/weak scaling, CMake, TotalView, Linaro DDT; Frontier (AMD MI250X), Perlmutter (NVIDIA A100), Frontera

ML and systems: PyTorch, PyTorch Geometric, JAX, equivariant GNNs, automatic differentiation, Docker, Kubernetes, Linux

Experience

PhD Researcher

Yale University, Electrical Engineering

2021–present

- Formulate and implement first-principles many-body models of self-trapped excitons and exciton-polarons in LiF and lead-free double perovskites.
- Maintain distributed Python/C++ simulation workflows using MPI, OpenMP, PETSc, and SLEPc for INCITE-supported workloads on Frontier and Perlmutter.
- Port multi-GPU numerical workloads across CUDA and ROCm; use scaling studies, collective communication, and TotalView/Linaro DDT to isolate distributed failures.
- Construct DeePH-style computational graphs and train equivariant graph neural networks with PyTorch Geometric and automatic differentiation to accelerate optical-property calculations; parts of the workflow achieved roughly 100x speedup by avoiding separate expensive derivative calculations.
- Co-author a 2025 *Journal of the American Chemical Society* paper and present related first-principles research at APS meetings.

Software and Embedded Systems Engineer

Probe Technologies / Weatherford Wireline Services, Houston, TX

2019–2021

- Implemented CUDA/OpenCL waveform-processing and OpenGL visualization tools for acoustic well-logging diagnostics.
- Delivered C# and ASP.NET services backed by Microsoft SQL Server for remote tool testing, sensor calibration, and engineering data management.

Selected Projects

Finite-element simulation portfolio — Implemented transient heat, Poisson, and incompressible-flow cases in DOLFINx/PETSc, with PyVista and Matplotlib post-processing.

EGNN force model — Trained an equivariant molecular model on QM9 using PyTorch Geometric/e3nn; differentiated predicted energies to obtain forces.

H2 VQE benchmark — Built a reproducible PySCF-to-Qiskit workflow with Jordan–Wigner mapping, UCCSD VQE, and exact-diagonalization reference curves.

Education

PhD, Electrical Engineering

Yale University, Computational materials, scientific computing, and ML

2021–present

MPhil and MS, Electrical Engineering

Yale University

2023

MEng and BS, Electrical and Computer Engineering

Cornell University,

2013–2018

Selected Publications and Presentations

Journal article: de Paula et al., including S. Vadivel, “Time-domain observation of ultrafast self-trapped exciton formation in lead-free double halide perovskites,” *JACS*, 2025.

APS presentations: First-principles studies of self-trapped excitons and exciton-polarons, 2024–2026.

Additional Technical Work

Containerized scientific software — Packaged CUDA-ready and cross-environment first-principles stacks with Docker and published reusable images to Docker Hub.